

Matthew Davis

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Education

University of Pittsburgh

B.S. in Computer Science — Frederick Honors College

Pittsburgh, PA

Expected May 2027

GPA: 3.93 / 4.00 — Accelerated 3-Year Completion

- **Relevant Coursework:** Data Structures and Algorithms 2, Natural Language Processing, Software Quality Assurance, Operating Systems, Databases, Computer Organization and Assembly Language, Formal Methods in Computer Science, Mathematical Foundations of Machine Learning, Linear Algebra

Experience

Vertex Inc

King of Prussia, PA (Remote)

Software Development Intern II

May 2026 – Aug 2026

- Developed a reusable SOAP-to-REST migration skill for modernizing enterprise connector APIs.
- Contributed to modernization initiatives including a shared File Share client library, VIPER-based CI/CD, legacy repository parity resolution, and a caching service for tax jurisdiction data.
- Applied Agentic Engineering and RPI (Research, Plan, Implement) methodology with GitHub Copilot and Claude to research legacy systems, design solutions, implement changes, and validate functionality.

Software Development Intern

May 2025 – Aug 2025

- Implemented role-based access control and secured APIs for the VAT Compliance Metrics Dashboard.
- Built backend API features for submission rates and workflow status metrics with advanced filtering.
- Improved CI/CD and cloud infrastructure using AWS, Terraform, and Route53 while supporting UI fixes and database migrations.

IT Intern

June 2024 – Aug 2024

Bethlehem Area School District

Bethlehem, PA

- Deployed 1,200+ devices across 22 schools serving 13,000 students, including system integration and reimaging.
- Diagnosed and repaired 300+ devices from a 20,000-device inventory for repair, reuse, and parts recovery.

Projects

Personality Space Explorer | Python, React, Vite, Three.js, scikit-learn, Pandas, UMAP, HDBSCAN

- Built an interactive 3D personality explorer using UMAP embeddings and HDBSCAN clustering across 50K+ personality profiles.
- Developed a Python ML pipeline to generate embeddings and Parquet artifacts, visualized through an interactive Three.js point cloud.

CozyNest RAG Pipeline | Next.js, TypeScript, PostgreSQL, Supabase, pgvector, Gemini, Groq

- Built an end-to-end RAG pipeline to ingest, chunk, embed, and retrieve support documents using PostgreSQL and pgvector similarity search.
- Developed a grounded AI support assistant that retrieves relevant context and generates responses with source citations and similarity scores.

Wordscope Extension | React, TypeScript, Plasmo, Stripe API, OpenAI API, Vercel, HTML/CSS

- Built a cross-browser word lookup extension integrating 7+ dictionary APIs with persistent history and data export.
- Integrated GPT-powered contextual explanations and token-based subscriptions using OpenAI and Stripe APIs.

Technical Skills

Languages: Python, Java, C++, C#, SQL, JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries: React, Next.js, Spring Boot, Node.js, Pandas, Three.js, scikit-learn

Tools & Platforms: AWS, PostgreSQL, Supabase, pgvector, Terraform, GitHub Actions, Vercel, Vite, Plasmo, OpenAI API, GitHub Copilot, Claude

Concepts: REST & SOAP APIs, RAG, Agentic Engineering, CI/CD, API Design & Integration, Machine Learning, Dimensionality Reduction, Clustering